

Technical Note

BULK HANDLING PROCEDURES - TYZOR® TPT

DESCRIPTION

Dorf Ketal Tyzor® organic titanates comprise a series of titanium and zirconate esters ranging from very reactive Tyzor® TPT (tetraisopropyl titanate) to stabilized chelated forms such as Tyzor® LA, a lactic acid chelate. Members of this titanate series vary in stability and solubility characteristics, as well as reactivity.

Titanates are proving useful in a wide range of industries. For example, they are used as polymerization catalysts with polyester, phenolic, silicone and epoxy resins. In these applications, titanates provide rapid cures at relatively low temperatures and often improve hardness, solvent resistance and electrical properties.

Titanates will hydrolyze to deposit a thin film of titanium dioxide on a variety of surfaces. In different applications, these thin films of titanium dioxide can impart adhesion-promotion effects to foils and films, scratch resistance on glass, improved electrical properties on electronic components, and aid in pigment dispersion.

Bulk handling of these products is frequently attractive, and this document provides general information concerning the bulk storage and handling of Tyzor® TPT.

SHIPPING CONTAINERS

Phosphated, unlined steel drums are normally used for small unit shipments. These must be purged with dry air or nitrogen before loading because atmospheric humidity will cause product hydrolysis.

TYZOR® TPT ORGANIC TITANATE

There is no unusual chemical hazard in handling Tyzor® TPT. Physical and chemical properties related to pumping and storing these materials are as follows:

1. Non-corrosive.
2. Reacts with water
3. Freezes at <18 - 20 °C (64 - 68 °F)
4. Viscosity increases as temperature declines
5. Color increases as storage temperature increases

TRUCK TRAILERS AND TANK CARS

Tanks must be clean and completely free of moisture prior to filling and should not be allowed to breathe atmospheric moisture during shipment. It is recommended that dry air or dry nitrogen be added as displacement during unloading to avoid moisture contamination. Empty trailers can be rinsed with an anhydrous alcohol solvent, then cleaned by conventional means.

If Tyzor® TPT residue is exposed to moisture prior to being rinsed out, a very stable layer of titanium dioxide will be deposited on tank surfaces.

This deposit cannot be removed by dilute acid, alkali or solvent. Tyzor® TPT forms very stable long-chain linkages in the early stages of hydrolysis. In time the hydrolysis will be complete and the material can be mechanically brushed off dry or with the aid of soapy water and scouring powder. The process of hydrolysis can be speeded up by application of hot water or steam. Successive scrubblings are more effective than a single exhaustive attempt.

- Steel trailers are not recommended due to cleaning and contamination problems.

Stainless Steel: Type 316 is an ideal material of construction. Lower grade alloys are less desirable because rust bloom from previous use with another material may cause color contamination.

- Glass or Heresite-lined steel are satisfactory but provide poor resistance to mechanical damage.

STORAGE TANKS

Tanks must be clean and blanketed with dry air or nitrogen. A static air or nitrogen pad of two or three ounces is preferred to a purge system due to the cost of nitrogen and the difficulty of obtaining a source of dry air.

One pound of water will degrade about 30 pounds of tetraisopropyl titanate to a binary hydrolysis product. Air or nitrogen used for padding a storage tank should therefore have a dew point of no more than -17.8 °C (0 °F) but preferably -40 °C (-40 °F) in order to minimize product loss.

Tyzor® TPT temperature should be maintained at about 20 °C (68 °F) to avoid problems with high viscosity or solidification. Temperature-controlled steam tracing or electric heating cable can be wrapped around tanks to supply heat. An external heat exchanger and appropriately designed circulating pumps can be used to maintain storage tank temperature.

1. Steel tanks can be used, but are not recommended due to problems with rust contamination and cleaning.
2. Stainless Steel Type 316 is very good. Type 304 is fairly satisfactory, but is subject to rust bloom.
3. Glass or Heresite-lined steel are satisfactory but are subject to mechanical damage.
4. Polyester and epoxy resins reinforced with glass fiber provide the best combination of economy, durability and chemical inertness for tanks and tank linings.

PUMPS AND VALVES

When Tyzor® TPT is allowed to contact moist air, it hydrolyzes to form solid titanium dioxide and long-chain intermediate compounds that will freeze pump shafts and destroy seals and packing.

Rotating shafts exposed to the liquid phase (e.g., horizontal centrifugal pumps) should be equipped with double mechanical seals and external circulating seal flush systems such as "Circuit Pak." The flush liquid can be isopropyl alcohol or any suitable organic liquid that is compatible with process materials. Water cannot be used as a flush liquid.

Centrifugal open-shaft or canned pumps are satisfactory for transferring Tyzor® TPT. If material is to be handled at temperatures below 20 °C (68 °F), special design considerations will be needed for pumping high-viscosity fluids.

PIPING

Stainless steel is recommended for piping. Heat and insulation are needed for outside piping and frequently for inside piping when handling Tyzor® TPT. Steam tracing should be sandwiched between two layers of insulation to avoid discoloration of material by overheating. Teflon® TFE fluorocarbon resin insulated electric heating cable is more easily installed than steam tracing and may be placed in contact with the piping. Thermo-switches should be installed at appropriate spots to control pipe temperature.

LUBRICANTS

Some care must be used in selecting lubricants that come in contact with titanates. Sulfur, for example, will develop color. Glycol-based greases will react to form elastomeric or rigid polymers. High-performance lubricants such as Krytox® fluorinated lubricants can be used for lubricating valve stem packing, etc. Krytox® is ideal because it is totally inert.

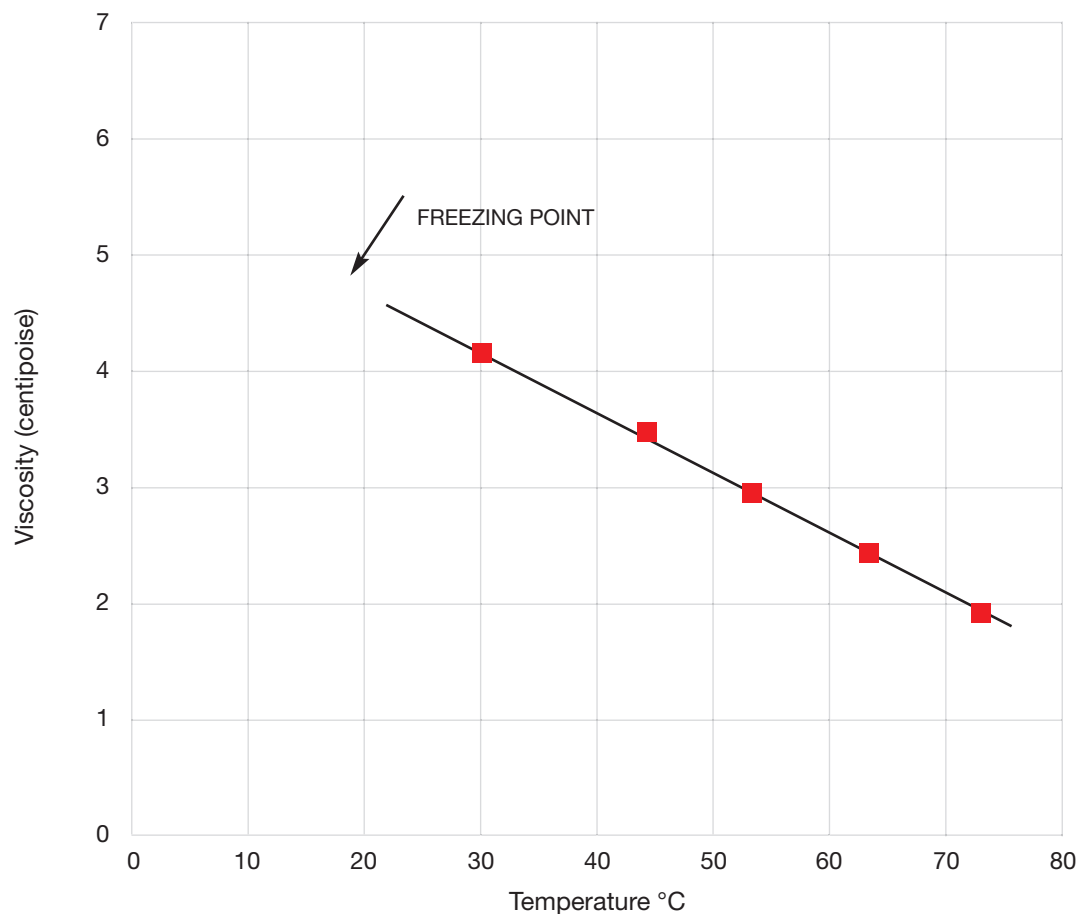
Physical and Chemical Properties

PROPERTY	TYPICAL VALUE
Formula	Ti(OC ₃ H ₇) ₄
Molecular weight	284
Melting point	17 - 19 °C (62.6 - 66.2 °F)
Boiling point	232 °C @ 760mm Hg (449.6 °F @ 29.93 in Hg) 03 °C @ 10mm Hg (37.4 °F @ 0.394 in Hg)
Vapor pressure @ 107 °C (224.6 °F)	10.0mm Hg (0.394 in Hg)
Viscosity @ 25 °C (77 °F)	4.5 centipoise
Density, liquid @ 25 °C (77 °F)	.955 g/cc (0.035 lb/cu in)
Heat capacity	0.82 PCU/lb °C @ 50 °C (0.82 BTU/lb °F @ 122 °F) 0.95 PCU/lb °C @ 150 °C (0.95 BTU/lb °F @ 302 °F)
Heat of vaporization @ 140 °C (284 °F)	-44 PCU/lb °C (-44 BTU/lb °F)
Boiling point of major hydrolysis product	82.5 °C (180.5 °F) (Isopropanol)
Flash point Tyzor® TPT, as produced*	> 70 °C (158 °F)
Flash point when exposed to moisture	11 - 25 °C (51.8 - 77 °F)
Suggested solvents	Isopropanol, n-Heptane

* Tyzor® TPT should be considered flammable because it reacts with moisture in the air to produce isopropanol, which is flammable. In extreme cases where large quantities of water are present, the flash point may decline below 37.8 °C (100 °F). Please consult MSDS for fire-fighting procedures.

Tyzor® TPT (Dry) is not corrosive.

Note: 1 PCU/°C = 1 BTU/°F



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